# SCOPE×COS-B-Series

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## Chapter 0: Introduction

The standalone COS B-Series paper (hereafter, the B-Series) organized the process from interpretation to response generation regarding another person's statement into three pathways: B1 (Literal Meaning), B2 (Context), and B3 (Intent). This paper does not restate the definitions of the B-Series. For the basic definitions of B1, B2, and B3, readers are referred to the standalone B-Series paper.

This paper primarily refers to the standalone B-Series paper and the integrated SCOPE paper. When confirmation of the definition of an individual layer is necessary, the corresponding individual SCOPE layer paper is referenced. For L8, this paper follows the definition reorganized in v2. These materials are available on LinkedIn and GitHub.

SCOPE is a cognitive coordinate system for estimating, using observable output as a clue, what kinds of cognitive structures may have been involved behind that output. Although the B-Series and SCOPE may address the same output, their roles differ. The B-Series organizes the pathways through which a statement is interpreted and a response is generated. SCOPE examines what kinds of cognitive structures may have been involved in the background of an observed output or in the background of an interpretation based on that output. The two are not identical and do not correspond to each other on a one-to-one basis.

In particular, similarities in names or apparent functions may lead to readings in which B1 is mapped to L1, B2 to L2, and B3 to L8. This paper does not adopt such mappings, for the following reasons.

- L1 is the verbalization layer that transforms cognitive content into a shareable form. B1, by contrast, is an interpretive pathway that uses the meanings and definitions of words as its primary basis.
- L2 is a cognitive coordinate representing emotional configuration toward a target and is not a function that performs processing. B2, by contrast, is an interpretive pathway that

proceeds by referring to factors such as situation, emotion, conversational flow, and interpersonal relationships.
- L8 is an accumulative foundation for materials available to cognition and the relational structures among them, and is not itself an interpretive pathway.

In none of these cases does similarity in name or apparent function constitute evidence for a correspondence.

This paper also does not claim that entities, functions, or processing programs corresponding directly to B1 through B3 or to individual SCOPE layers exist as such within humans or LLMs. These are descriptive frameworks for organizing and comparing observed outputs and phenomena in a consistent form.

The purpose of this paper is to examine what kinds of connection hypotheses can be placed between the interpretive pathways organized as B1, B2, and B3 and SCOPE. Response generation itself, although included within the scope of the B-Series, is not a central object of examination in this paper.

A single interpretive pathway may hypothetically relate to multiple SCOPE layers. The same SCOPE layer may also be involved in multiple interpretive pathways. The connections presented in this paper do not imply that a particular layer must always be involved in a particular interpretive pathway. Nor do they imply that a particular cognitive structure can be uniquely determined from an observed output.

In B3, multiple hypotheses concerning the purpose of an utterance are generated, retained, and updated. In SCOPE, multiple structural hypotheses are retained while estimation proceeds using observed output as a clue. Both deal with multiple hypotheses, but the objects of those hypotheses differ.

A B3 hypothesis concerns “what intent may lie behind this utterance.” A SCOPE hypothesis concerns “what kinds of cognitive structures may have been involved in the background of an observed output, or in the background of an interpretation based on that output.” Hypotheses concerning the background of an output and hypotheses concerning the background of an interpretation are treated as separate hypotheses. Neither is a hypothesis about the intent behind the utterance. This paper maintains these distinctions throughout the discussion.

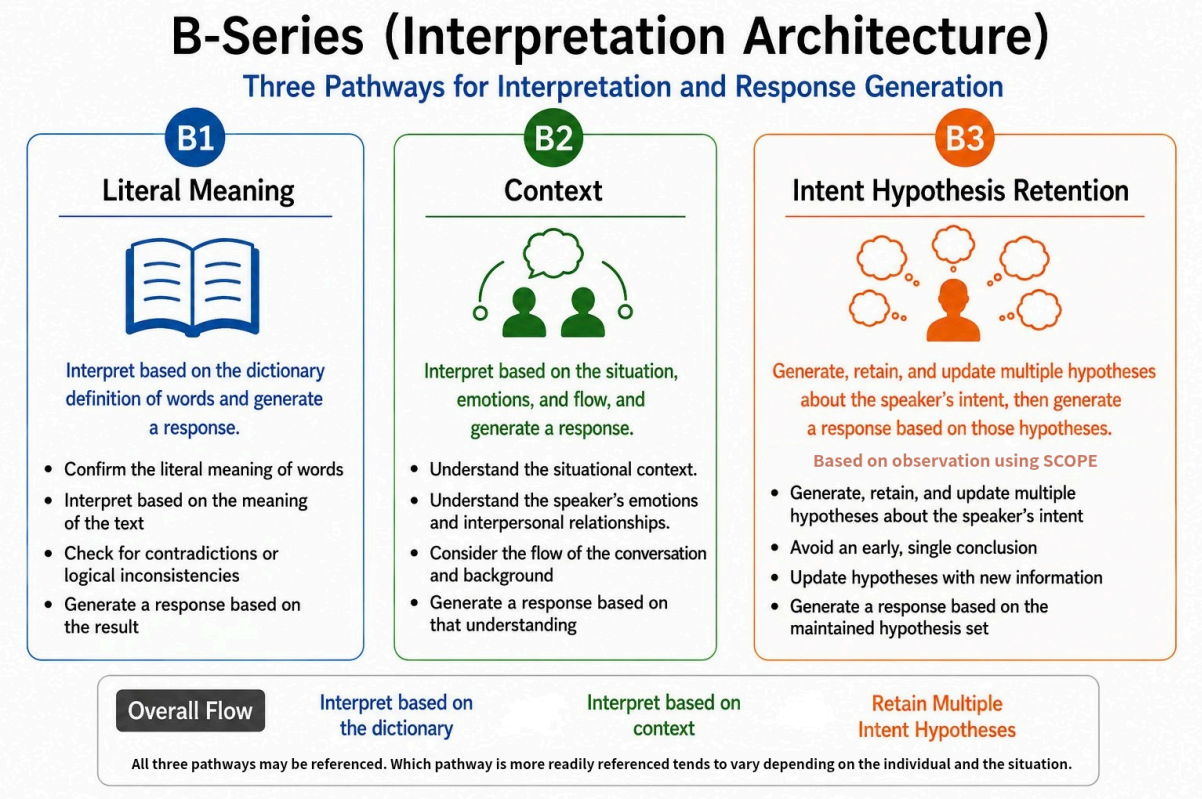
This paper primarily treats a single output that has become observable in linguistic form as the unit of analysis. This does not redefine the unit of analysis of the B-Series itself; it is a scope limitation adopted in this paper for examining the connection between B and SCOPE.

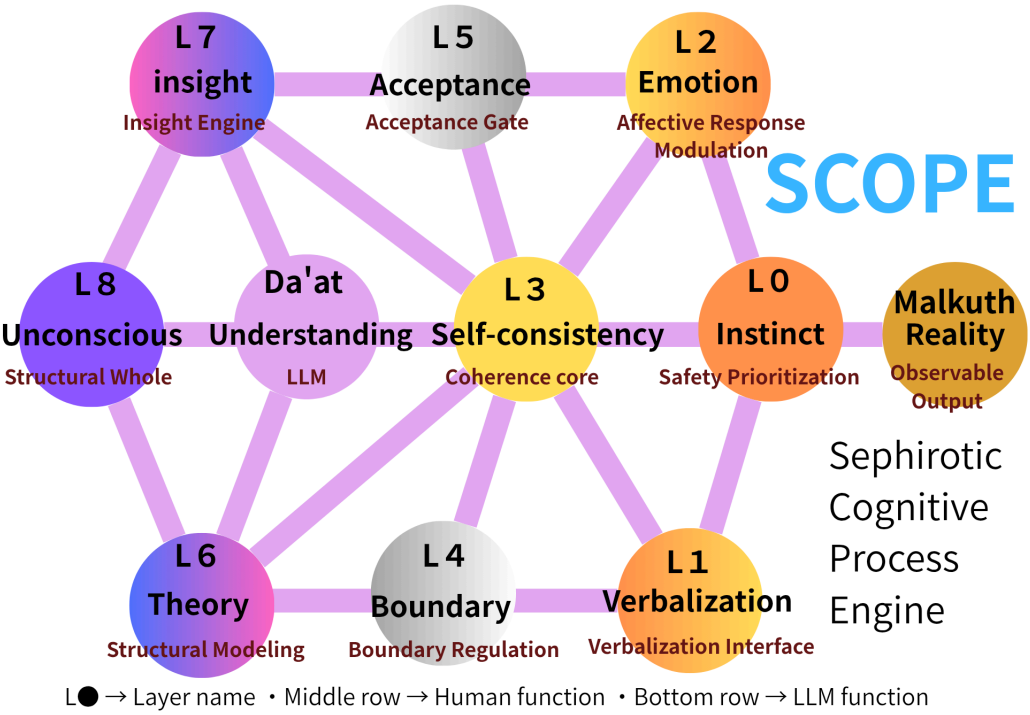
Unverbalized internal cognition and non-linguistic cognition itself are not direct objects of analysis. This does not mean that the paper is limited to cognitive agents that primarily think in language. Rather, regardless of the nature of the cognitive agent, this paper addresses output at the point at which cognition has been expressed in linguistic form.

Under this scope, the paper primarily considers two cases: a human interpreting another person’s linguistic output, and an LLM interpreting a user’s linguistic output. In both cases,

the interpreting agent and the agent that produced the output are different. The application of this framework to self-analysis of one's own linguistic output or internal cognition is outside the scope of this paper.

On this basis, the following chapters examine B1, B2, and B3 in sequence and consider what kinds of hypothetical relationships may be placed between each interpretive pathway and SCOPE.



# Chapter 1: B1 — Literal Meaning and SCOPE

In the standalone B-Series paper, B1 is organized as an interpretive pathway that uses the meanings and definitions of words as its primary basis. This chapter examines what kinds of connection hypotheses can be proposed between this pathway and SCOPE.

Using literal meaning as the primary basis does not mean that interpretation must converge on a single literal meaning. When a word or abbreviation has multiple meanings in a language, multiple interpretations may be possible even at the level of literal meaning alone.

Consider the linguistic output, “Could you handle the PR?” The abbreviation “PR” may refer to public relations, but in software development it may also refer to a pull request. In settings such as a workplace where public relations staff and software developers work together, either interpretation may be possible at the level of literal meaning. This example is provided for explanatory purposes and does not serve as evidence for the argument of this chapter. How reference to the situation may change the interpretation will be addressed in Chapter 2.

When literal meaning and the observed output alone are sufficient to organize what is needed for interpretation, there is no need to introduce an additional structural hypothesis through SCOPE. In the PR example, if it is sufficient to organize the output as allowing two interpretations at the level of literal meaning, no further hypothesis may be necessary.

Is it then necessary to examine why the interpreting agent interpreted the output according to one meaning rather than the other? A single linguistic output does not by itself establish which meaning was used in the output or which meaning was used in interpretation. When literal meaning can be clarified through subsequent dialogue, a divergence in literal meaning may be resolved. However, the exchange used for clarification is itself subject to interpretation, and clarification does not necessarily resolve the divergence. In some cases, resolving a divergence in literal meaning requires only confirming which meaning was used in the output and which meaning was used in interpretation, without examining why that particular meaning was taken. Within this scope, it may therefore be unnecessary to introduce an additional structural hypothesis concerning why that meaning was used in interpretation. The decision of whether to seek clarification is not addressed in this paper.

Even when an output is interpreted according to only one meaning, observed output alone cannot distinguish whether the other meaning was unavailable to the interpreting agent, whether it was not referenced at that time, or whether this reflects some other background. Accordingly, this chapter does not conclude, on the basis of a single literal interpretation alone, that the other meaning was unavailable to the interpreting agent, or draw similar conclusions. This is consistent with the SCOPE premise that the cognitive structure behind an observed output cannot be uniquely determined from that output.

The same organization can be applied at the level of literal meaning when an LLM interprets a user’s linguistic output. If a user produces the output, “Could you handle the PR?”, both meanings may be possible at the level of literal meaning. Even if the LLM’s response follows

one of those meanings, that response alone cannot establish whether the other meaning was referenced or used, nor can it establish why the response followed one meaning rather than the other. A divergence in literal meaning may also be resolved through clarification in subsequent dialogue.

Within the scope examined in this chapter, the connection between B1 and SCOPE can therefore be organized as follows. When a divergence in literal meaning can be resolved through clarification of literal meaning in subsequent dialogue, it may be unnecessary to introduce an additional structural hypothesis through SCOPE. This does not prohibit placing SCOPE hypotheses concerning B1. Nor does it establish a correspondence between B1 and any particular SCOPE layer.

The next chapter examines the connection between SCOPE and B2, the interpretive pathway that references factors such as the situation and the flow of conversation.

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# Chapter 2: B2 — Context and SCOPE

In the standalone B-Series paper, B2 is organized as an interpretive pathway that interprets and generates responses based on situation, emotion, and conversational flow, with the atmosphere of the setting, emotional tone, and interpersonal relationships, among others, as its objects of reference. This chapter focuses on the aspect of B2 that references situation and conversational flow, and examines what kind of connection hypothesis can be placed between this pathway and SCOPE. Other aspects are not addressed in this chapter. This does not change the definition of B2.

The unit of analysis in this paper is primarily a single linguistic output. However, the information referenced when interpreting that output may extend beyond the unit of analysis itself. Situation and conversational flow lie outside the linguistic output that constitutes the unit of analysis, yet may be referenced in its interpretation. This chapter addresses cases in which such situations or conversational flow outside the linguistic output are referenced. Cases in which other words or information contained within the same linguistic output make one interpretation more compatible are not addressed in this chapter.

Chapter 1 established that referencing literal meaning does not mean that interpretation is determined by a single literal meaning. When multiple interpretations are possible at the literal level, referencing only the meanings and definitions of words may not be sufficient to determine one interpretation. If, nevertheless, the output is interpreted in one of these meanings, something other than literal meaning may have been referenced in that interpretation. Situation and conversational flow are among the possible candidates.

Consider the single linguistic output, "Could you handle the PR?" As seen in Chapter 1, two interpretations may be possible at the literal level. If this output occurs in a conversational flow in which advertising or publicity has just been discussed, an interpretation of PR as public relations may be more compatible with that flow. If code changes have just been discussed, an interpretation of PR as a pull request may be more compatible. Likewise, if the output is interpreted in one meaning on the basis of information such as the output

producer's occupation or industry, this can be organized as a reference to the situation. This example is provided for explanatory purposes and does not constitute the basis of the argument in this chapter.

However, the fact that an interpretation is compatible with the situation or conversational flow does not show that the situation or flow was actually referenced in the interpretation. Whether the interpreting agent took one meaning by referencing the situation or conversational flow cannot be distinguished from the observed output alone, just as the other possibilities concerning the other meaning discussed in Chapter 1 cannot.

Candidate literal meanings can be identified from the words contained in the linguistic output that constitutes the unit of analysis. By contrast, for the situations and conversational flow addressed in this chapter, which lie outside the output, neither which of them was actually referenced nor whether the output producer and the interpreting agent referenced the same situation or conversational flow can be determined from that linguistic output alone. Furthermore, referencing situation or conversational flow does not guarantee that the interpretation is correct, nor does an interpretation formed through such reference necessarily match the meaning in which the output producer used the word. B3, addressed in the next chapter, is organized as a pathway that interprets while maintaining multiple hypotheses about the purpose for which the output producer produced the output.

Even so, as in Chapter 1, resolving a divergence in literal meaning may in some cases require no more than clarifying which meaning was used in the output and which meaning was interpreted. In such cases, it may not be necessary to determine which situation or conversational flow was referenced, and to that extent, an additional structural hypothesis may not be required. On the other hand, when asking which situation or conversational flow was referenced, or why it was utilized in the interpretation, a hypothesis about what was referenced must be distinguished from a hypothesis about why it was utilized in the interpretation. In addition, when placing a SCOPE hypothesis, it is necessary to distinguish whether that hypothesis concerns the background of the output or the background of the interpretation based on that output.

The same organization can be applied when an LLM interprets a user's linguistic output. If a user outputs "Could you handle the PR?", earlier exchanges may be referenced as conversational flow outside the linguistic output that constitutes the unit of analysis. Even if the LLM's response follows one meaning that is compatible with those earlier exchanges, that response alone cannot determine which exchanges were referenced, whether the other meaning was referenced or utilized, or why the response followed one meaning rather than the other.

Within the scope examined in this chapter, the connection between B2 and SCOPE can therefore be organized as follows. Even when an interpretation following one meaning is compatible with the situation or conversational flow, to the extent that the divergence in literal meaning can be resolved through clarification, there may be no need to add a structural hypothesis based on SCOPE. On the other hand, when asking what was referenced or why it was utilized in the interpretation, the observed output alone cannot provide an answer, leaving room to consider a structural hypothesis based on SCOPE when necessary. However, a structural hypothesis based on SCOPE does not itself determine which situation

or conversational flow was actually referenced. This neither makes it necessary to place a SCOPE hypothesis regarding B2, nor establishes a correspondence between B2 and any particular SCOPE layer.

The next chapter examines the connection between SCOPE and B3, an interpretive pathway that focuses on the purpose for which the output producer produced the output and maintains multiple hypotheses while interpreting it.

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# Chapter 3: B3 — Intent and SCOPE

In the standalone B-Series paper, B3 is organized as a pathway that, based on observation using SCOPE, interprets while generating and maintaining multiple hypotheses about communicative intention, that is, why the output producer produced the output, and generates a response. These hypotheses are not conclusively determined as a single answer, and are updated when new information becomes available. Unlike B1, discussed in Chapter 1, and B2, discussed in Chapter 2, B3 includes being based on observation using SCOPE in the definition of the pathway. Without changing this existing definition, this chapter examines what role SCOPE, as referenced in B3, plays. The subject of this paper is B3 within COS; inference of communicative intention conducted without using COS is not addressed in this paper.

What B3 maintains are multiple hypotheses concerning communicative intention, that is, why the output producer produced the observed output. What matters here is not determining these hypotheses as a single correct answer, but proceeding with interpretation while maintaining them as multiple possibilities. Because determining one hypothesis may remove other possibilities from those being maintained, B3 maintains multiple hypotheses. The purpose of B3 is not to correctly identify the output producer's communicative intention.

When the output producer explains their own intention, that explanation may serve as information for updating the hypotheses concerning communicative intention. However, the intention is not treated as conclusively determined on that basis alone. The explanation does not necessarily state the output producer's communicative intention accurately. Therefore, the output producer's own explanation is not treated as a condition for conclusively determining one hypothesis.

In this chapter, what SCOPE addresses is not the hypotheses about communicative intention maintained by B3 themselves, but structural hypotheses concerning what kind of cognitive structure may have been involved in the background of the observed linguistic output. SCOPE uses observable output as a clue to infer the cognitive structure behind it. Accordingly, structural hypotheses based on SCOPE do not conclusively determine the output producer's internal state or communicative intention.

The hypotheses concerning communicative intention maintained by B3 and the structural hypotheses concerning cognitive structure based on SCOPE are not the same. B3 maintains multiple possibilities concerning communicative intention, that is, why the output producer produced the output. SCOPE, by contrast, organizes what kind of cognitive structure may

have been involved in the background of that output as separate structural hypotheses. The difference in roles between the two models, with COS addressing cognitive processing and SCOPE inferring cognitive structure, is maintained in B3 as well.

Under the existing definition, B3 is based on observation using SCOPE. However, structural hypotheses based on SCOPE do not conclusively determine any of the hypotheses concerning communicative intention maintained by B3 as the correct answer. When new information becomes available, the hypotheses concerning communicative intention maintained in B3 are updated.

What is at issue in B3 is communicative intention, that is, why the output producer produced the output, and this differs in object from the reference to situation and conversational flow discussed in Chapter 2. However, this does not mean that interpretation through B2 is incorrect and interpretation through B3 is correct. Nor does it mean that using B3 makes it possible to accurately grasp the output producer's communicative intention or internal state.

In addition, the divergence in literal meaning discussed in Chapters 1 and 2 may in some cases be resolved through clarification. However, even if it is clarified which meaning was used in the output, or which situation or conversational flow was referenced, this does not necessarily determine the communicative intention. Clarification of meaning or of what was referenced must be distinguished from inference of communicative intention.

The same distinction is necessary when an LLM interprets a user's linguistic output. Even if multiple candidates for communicative intention are generated, there is no guarantee that any of them is the actual intention of the output producer. In addition, even if a response follows one communicative intention, that response alone cannot determine whether other candidates were maintained. Furthermore, the LLM's response alone cannot determine which candidate was referenced or utilized, or why that response was generated.

B3 covers interpretation up to response generation. The form in which the generated response is conveyed externally is an area addressed by the C-Series. This distinction follows the existing organization of the B-Series and C-Series.

Within the scope examined in this chapter, the relationship between B3 and SCOPE can therefore be organized as follows. Unlike B1 and B2, discussed in Chapters 1 and 2, B3 is organized in the standalone B-Series paper as a pathway of interpretation and response generation based on observation using SCOPE. This chapter does not change that definition. What this chapter has distinguished are the multiple hypotheses concerning communicative intention maintained by B3 and the structural hypotheses concerning cognitive structure based on SCOPE. Neither conclusively determines the output producer's communicative intention or internal state. Nor does this relationship imply a correspondence between B3 and any particular SCOPE layer.

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# Chapter 4: Limitations in the Connection between the B-Series and SCOPE

This chapter organizes the limitations that apply to the connection between the B-Series and SCOPE as a whole, based on the distinctions discussed in Chapters 1 through 3.

First, connecting B and SCOPE does not make it possible to uniquely determine the communicative intention or internal state behind an observed output. The hypotheses about communicative intention maintained in B3 and the structural hypotheses based on SCOPE both remain estimates, and this connection does not guarantee a correct answer.

Second, even when B and SCOPE are connected, it is not guaranteed that every possible communicative intention will be generated as a hypothesis in B3. B3 generates and maintains multiple hypotheses about communicative intention, but possibilities that are not conceived are not included among the maintained hypotheses, at least at that point in time. Therefore, when a discrepancy in interpretation occurs, its cause may lie not only in an error within the maintained hypotheses, but also in the absence of the candidate itself. A missing hypothesis candidate and an incorrect maintained hypothesis need to be treated as distinct problems.

Third, knowing or using the B-Series or SCOPE does not guarantee that the tendency regarding which pathway is most readily referenced will change. Nor does it guarantee that the pathway referenced will change when the situation changes. The standalone B-Series paper states that the three pathways do not represent levels of proficiency or a hierarchy of quality, and that the pathway most readily referenced tends to vary depending on the individual and the situation. The fact that one pathway is more readily referenced does not mean that the other pathways are not referenced.

This limitation also requires attention when considering LLMs. The standalone B-Series paper states that many current LLMs may possess processing that references the literal meaning and context of an utterance to generate a response. Within the B-Series framework, this corresponds to processing in the range of B1 through B2. Regarding the maintenance of multiple hypotheses about communicative intention, as addressed by B3, it remains unclear to what extent such processing is functioning. Even if providing the B-Series or SCOPE framework as input changes the output within a particular interaction, this alone does not mean that the underlying tendency regarding which pathways are referenced has changed.

Fourth, this paper has not empirically validated the connection between B and SCOPE organized here. Questions remain regarding how interpretive processes and cognitive structures can be handled in a verifiable form on the basis of observed output, as well as how the possible influence of presenting the framework itself on subsequent interpretation or responses should be handled in the design of validation procedures. Including these questions, further consideration is required regarding procedures for empirically examining the connection organized in this paper. The purpose of this paper is to organize the relationship between B and SCOPE theoretically; it does not empirically establish its validity.

In summary, connecting B and SCOPE does not determine communicative intention or internal state, does not guarantee a correct answer, and does not guarantee that every possible communicative intention will be generated as a hypothesis in B3. In addition, knowing or using the B-Series or SCOPE does not guarantee that the tendency regarding which pathway is most readily referenced, or the pathway actually referenced in a particular situation, will change automatically. Furthermore, the validity of the connection organized in this paper has not yet been empirically validated. This chapter has set out the limitations that remain even when B and SCOPE are connected.

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# Chapter 5: The Significance of Treating Intent Hypotheses and Structural Hypotheses Separately

Building on the limitations organized in Chapter 4, this chapter examines what significance there is, nevertheless, in distinguishing hypotheses about communicative intention from structural hypotheses based on SCOPE.

The hypotheses examined in this chapter are distinguished from the hypotheses maintained by the receiver at the time. Here, the party analyzing the observed output, following the distinctions of the B-Series and SCOPE, places hypotheses about communicative intention and hypotheses about cognitive structure. This is not an analysis by a party to that output of their own internal cognition. Even if the receiver did not maintain hypotheses about communicative intention, hypotheses about communicative intention can still be examined analytically. However, this does not indicate which interpretive pathway the receiver actually referenced, nor does it mean that the receiver referenced B3.

This chapter examines discrepancies in interpretation from the perspective of incompatibility between hypotheses about the meaning or communicative intention with which the output producer produced a given linguistic output and hypotheses about the receiver's interpretation of that same output. This does not mean that the meaning or communicative intention actually used by the output producer, or the receiver's interpretation, is given as an observed fact.

In this examination, the following distinctions are made regarding where a discrepancy may possibly lie. One is the level of reference to literal meaning or situation, discussed in Chapters 1 and 2. Another is the level of hypothesis types, at which hypotheses about communicative intention, structural hypotheses about the background of the output, and structural hypotheses about the background of the interpretation based on that output are treated separately. Furthermore, when the analyzing party assumes that multiple hypotheses about communicative intention were maintained on the receiver's side, the possibility that another candidate was not included among them may also be examined. However, this possibility can only be examined retrospectively, when the analyzing party later considers another candidate. All of these descriptions remain estimates and do not establish the actual cause of the discrepancy.

If these are not distinguished, a discrepancy may be reduced to a single "interpretive error." In addition, one type of hypothesis may be treated as establishing another type. Examining the relationship between the two types of hypotheses is not itself the problem. The problem is treating hypotheses about communicative intention and structural hypotheses circularly as grounds for one another, and regarding this alone as providing observational support. In such a case, both hypotheses may continue to be maintained without being examined against observed output.

Describing the location of a discrepancy is carried out by returning to observed output. Distinguishing types of hypotheses does not increase the observed output. An output in which the receiver expresses in language how they interpreted that output may become a separate observed output for examining hypotheses about the receiver's interpretation and structural hypotheses about the background of that interpretation. Such a subsequent output may serve as material for examining hypotheses about the preceding output, but it is itself treated as a separate unit of analysis. In addition, the receiver's statement of their interpretation is not used as material for establishing the output producer's communicative intention.

When a response is observed, whether that response contains the receiver's interpretation is treated as a separate question. Even if a response follows one of the meanings, if it does not contain the receiver's interpretation, how the receiver interpreted the output, or whether other candidates were referenced, cannot be identified from that response alone. Even when an output containing the receiver's interpretation is observed, this guarantees neither that the location of the discrepancy will be established nor that the discrepancy will be resolved. What is obtained is material for examining the location of the discrepancy.

An output in which the receiver presents their own interpretation and then checks whether it is correct may arise from different interpretive pathways. It may be a case in which an interpretation through B2 is presented in a form that can be checked rather than as settled, or the same form of output may arise while multiple hypotheses about communicative intention are maintained in B3. Which pathway was referenced cannot be determined from the form of the observed output. Describing the location of a discrepancy separately by hypothesis type does not depend on which pathway the receiver referenced. At the same time, such a description presupposes the use of the distinctions of the B-Series and SCOPE. This does not mean that B3 is superior to B2.

When an LLM interprets a user's linguistic output, the LLM's response may contain its own interpretation in a form that can be checked. From the user's side, such a response becomes an output containing how the LLM interpreted that output, and may serve as material for examining the location of a discrepancy. However, from that response alone, it cannot be identified whether multiple candidates are maintained within the LLM or which pathway was referenced.

In summary, one significance of distinguishing hypotheses about communicative intention from structural hypotheses lies not in reaching a correct interpretation, but in making it possible, when a discrepancy in interpretation occurs, to describe where that discrepancy is being examined as possibly lying, as distinct, multiple hypotheses. Such a description does not establish the actual cause. An output containing the receiver's interpretation may serve

as material for that description. This description does not depend on which interpretive pathway the receiver referenced, but it presupposes the use of the distinctions of the B-Series and SCOPE.

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# Conclusion

This paper examined the connection between the interpretive pathways organized as B1, B2, and B3 and SCOPE. B1 is an interpretive pathway that uses the meanings and definitions of words as its primary basis, and B2 is an interpretive pathway that proceeds by referring to factors such as situation, emotion, conversational flow, and interpersonal relationships. In the examination of this paper, to the extent that a divergence in literal meaning can be resolved through clarification, there may be no need, for both B1 and B2, to add a structural hypothesis based on SCOPE. B3, by contrast, is a pathway that interprets while maintaining multiple hypotheses about communicative intention, that is, why the output producer produced the output. In the standalone B-Series paper, being based on observation using SCOPE is included in the definition of B3. This difference does not represent a hierarchy of quality or levels of proficiency among the pathways. It should be noted that the B-Series covers the process from interpretation to response generation, and that in B3, responses are generated on the basis of the multiple hypotheses maintained. However, response generation itself is not a central object of examination in this paper.

In addition, this paper distinguished hypotheses about communicative intention from structural hypotheses based on SCOPE as different types of hypotheses. If these are not distinguished, a discrepancy in interpretation may be reduced to a single error. One significance of treating them separately lies not in reducing a discrepancy to a single error, but in making it possible to describe where that discrepancy is being examined as possibly lying, as distinct, multiple hypotheses. Such a description remains an estimate.

However, connecting B and SCOPE does not determine communicative intention or internal state, nor does it guarantee a correct answer. This paper has organized the relationship between B and SCOPE theoretically; it has not empirically established its validity. Accordingly, the connection between B and SCOPE may be positioned as one framework for distinguishing and examining different types of hypotheses.